

Large In-Plane and Vertical Piezoelectricity in Janus Transition Metal Dichalcogenides

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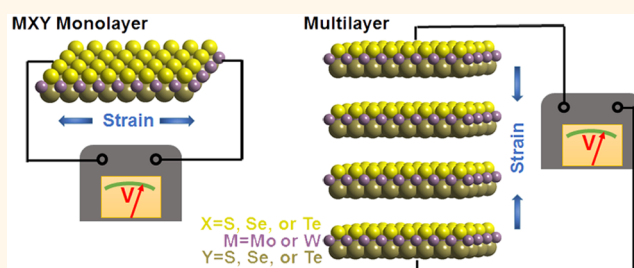
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S Supporting Information

ABSTRACT: Piezoelectricity in 2D van der Waals materials has received considerable interest because of potential applications in nanoscale energy harvesting, sensors, and actuators. However, in all the systems studied to date, strain and electric polarization are confined to the basal plane, limiting the operation of piezoelectric devices. In this paper, based on *ab initio* calculations, we report a 2D materials system, namely, the recently synthesized Janus MXY (M = Mo or W, X/Y = S, Se, or Te) monolayer and multilayer structures, with large out-of-plane piezoelectric polarization. For MXY monolayers, both strong in-plane and much weaker out-of-plane piezoelectric polarizations can be induced by a uniaxial strain in the basal plane. For multilayer MXY, we obtain a very strong out-of-plane piezoelectric polarization when strained transverse to the basal plane, regardless of the stacking sequence. The out-of-plane piezoelectric coefficient d_{33} is found to be strongest in multilayer MoSTe (5.7–13.5 pm/V depending on the stacking sequence), which is larger than that of the commonly used 3D piezoelectric material AlN ($d_{33} = 5.6$ pm/V); d_{33} in other multilayer MXY structures are a bit smaller, but still comparable. Our study reveals the potential for utilizing piezoelectric 2D materials and their van der Waals multilayers in device applications.

KEYWORDS: piezoelectricity, 2D materials, Janus transition metal dichalcogenides, monolayers, multilayers



The piezoelectric effect is an intrinsic electromechanical coupling between stresses/strains and electric polarizations/fields in semiconductors and insulators whose crystal structures lack inversion symmetry. Conventional piezoelectric materials, including α -quartz, II–VI and III–VI semiconductors (e.g., ZnO and AlN), perovskites (e.g., BaTiO₃ and PbTiO₃), etc., have found widespread applications as sensors, actuators, and power transducers/sources in electronic and medical industries.^{1–3} In fact, out of the total 32 classes of crystal structures, 20 classes can potentially display piezoelectricity, providing ample opportunities to discover other piezoelectric materials systems with potential device applications.

There has been a growing interest in recent years in looking for two-dimensional (2D) piezoelectric materials, to miniaturize the electromechanical devices and to exploit the excellent mechanical and electronic properties of 2D materials. In-plane piezoelectricity had been previously predicted in monolayer BN⁴ and transition metal dichalcogenides (TMDs) such as MoS₂⁵ using density functional theory (DFT) simulations. It was later confirmed by experiments that piezoelectric responses can indeed be obtained from stretched MoS₂ monolayers on substrates or by indenting free-standing MoS₂ monolayers.^{6,7} Both theoretical and experimental studies suggested that TMD monolayers possess strong piezoelectric coefficients that are

comparable and even larger than those of bulk α -quartz and II–VI and III–VI semiconductors,^{5,7} implying their potential in electronic and energy applications. Furthermore, it had been demonstrated that piezoelectric fields in MoS₂ monolayer and odd-numbered multilayer structures, coupled with their exciton properties, give rise to modulated photoluminescent strengths, which may find more optoelectronic applications such as optical delays.⁸ These studies triggered an intense interest in piezoelectric properties of other 2D materials, including 2D TMDs beyond the Mo and W families, 2D II–VI and III–V structures (e.g., ZnO and AlN monolayers), and group III and group IV monochalcogenides (e.g., GaS and SnSe).^{9–11} We also noted that the piezoelectric properties of one-dimensional (1D) nanotubes have also been investigated using theoretical approaches.⁴

One common feature of 2D piezoelectric materials studied so far is that their piezoelectric polarizations are confined within the basal plane, excluding the out-of-plane operation mode in devices based on these materials. Here we report a theoretical study on the basis of DFT simulations to reveal both in-plane

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and out-of-plane piezoelectric effects in Janus MXY (M = Mo or W, X/Y = S, Se, or Te, and X ≠ Y) TMD monolayers and their multilayer structures. These materials are semiconductors based on prior first-principles simulations.¹² Very recently, an example of such Janus MXY monolayers—MoSSe—has been successfully synthesized in one of the authors' lab by selective sulfurization of the top atomic layer in MoSe₂ at 800 °C while retaining the Se atoms in the entire bottom atomic layer.¹³ The structure of the MoSSe monolayer (Figure 1a) is similar to the

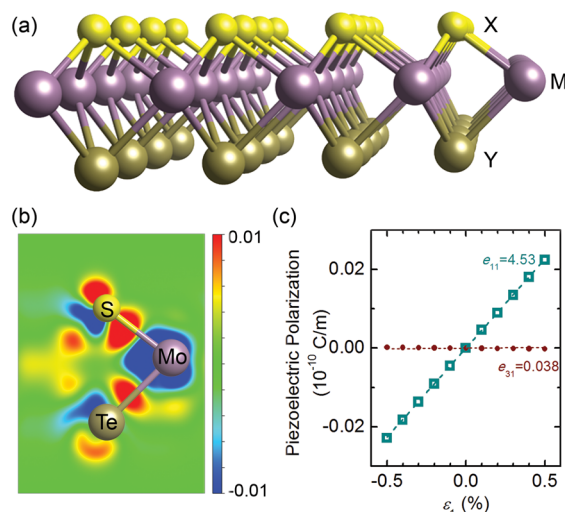


Figure 1. (a) Crystal structure of Janus MXY (M = Mo or W, X/Y = S, Se, or Te, and X ≠ Y) monolayers, with the yellow, purple, and brown spheres representing M, X, and Y elements, respectively. (b) Bonding charge density of the Janus MoSTe monolayer, obtained from the difference between the valence charge density of the monolayer and the superposition of the valence charge density of the neutral constituent atoms. Red and blue colors indicate the electron accumulation and depletion, respectively, with a scale unit e/Bohr^3 . (c) Linear changes in in-plane and out-of-plane piezoelectric polarizations of the MoSTe monolayer occur when subject to a uniaxial strain ϵ_1 between −0.5% and 0.5%, giving its e_{11} and e_{31} values (unit: 10^{-10} C/m).

regular MX₂ type (e.g., MoS₂) of monolayers in the 2H phase, as confirmed by transmission-electron microscopy measurements, Raman spectrum, and DFT simulations.¹³ But this monolayer is different from MoS_xSe_{1-x} solid solutions in the sense that S and Se atoms are never coplanar in the Janus MoSSe monolayer. In comparison to MX₂, the Janus TMD monolayers lack the reflection symmetry with respect to the central metal atoms, allowing out-of-plane electric polarizations. Our DFT simulations show that in Janus MXY monolayers subject to a uniaxial in-plane strain, an in-plane piezoelectric effect as strong as that in conventional MX₂ monolayers is generated; the out-of-plane piezoelectric effect however is much weaker. Strikingly, in multilayer structures, the out-of-plane piezoelectric effect is generated for strains applied in the direction of stacking regardless of the stacking sequence. The magnitude of this effect is comparable to and even stronger compared to AlN, the most frequently used bulk and thin film piezoelectric material. These results, together with the fact that a MoSSe monolayer has been synthesized, demonstrate the great potential of these Janus TMDs for nanoscale electronic and energy applications.

THEORY OF PIEZOELECTRICITY IN JANUS TMDS

The piezoelectric effect is an electromechanical coupling described by third-rank tensors e_{ijk} and d_{ijk} as

$$e_{ijk} = \partial P_i / \partial \epsilon_{jk} = \partial \sigma_{jk} / \partial E_i \quad (1)$$

and

$$d_{ijk} = \partial P_i / \partial \sigma_{jk} = \partial \epsilon_{jk} / \partial E_i \quad (2)$$

where P_i , E_i , ϵ_{jk} , and σ_{jk} represent piezoelectric polarizations, macroscopic electric fields, strains, and stresses, respectively. The subscripts i , j , and $k \in \{1, 2, 3\}$, where 1, 2, and 3 correspond to the spatial x - (armchair), y - (zigzag), and z - (vertical) directions, respectively. In the contracted Voigt notation, e_{ijk} and d_{ijk} are reduced to e_{il} and d_{il} , respectively, where $l \in \{1, 2, \dots, 6\}$. The number of independent components in e_{il} and d_{il} tensors is reduced by the crystal symmetry.¹⁴ Both the conventional MX₂ monolayers and the Janus MXY monolayers, as well as their multilayer structures, at least possess the 3m point group symmetry, giving

$$e_{il} = \begin{pmatrix} e_{11} & -e_{11} & \cdot & \cdot & e_{15} & \cdot \\ \cdot & \cdot & \cdot & e_{15} & \cdot & -0.5e_{11} \\ e_{31} & e_{31} & e_{33} & \cdot & \cdot & \cdot \end{pmatrix} \quad (3)$$

and

$$d_{il} = \begin{pmatrix} d_{11} & -d_{11} & \cdot & \cdot & d_{15} & \cdot \\ \cdot & \cdot & \cdot & d_{15} & \cdot & -2d_{11} \\ d_{31} & d_{31} & d_{33} & \cdot & \cdot & \cdot \end{pmatrix} \quad (4)$$

where the dot means 0. In this study, we perform DFT simulations to calculate the values of e_{il} and derive the values of d_{il} using the relation

$$e_{il} = d_{ik} C_{kl} \quad (5)$$

where C_{kl} is the elastic stiffness tensor

$$C_{kl} = \begin{pmatrix} C_{11} & C_{12} & C_{13} & C_{14} & \cdot & \cdot \\ C_{12} & C_{11} & C_{13} & -C_{14} & \cdot & \cdot \\ C_{13} & C_{13} & C_{33} & \cdot & \cdot & \cdot \\ C_{14} & -C_{14} & \cdot & C_{44} & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & C_{44} & C_{14} \\ \cdot & \cdot & \cdot & \cdot & C_{14} & 0.5(C_{11} - C_{12}) \end{pmatrix} \quad (6)$$

for the 3m point group symmetry. Equations 3 to 6 will be applied to the monolayer and multilayer structures separately in the next section. Note that the units of e_{il} , d_{il} , and C_{kl} are 10^{-10} C/m, pm/V, and N/m, respectively, in the 2D limit, whereas in 3D structures their units are C/m², pm/V, and 10^{10} Pa, respectively.

RESULTS AND DISCUSSION

MX₂ Monolayers. For 2D semiconductors, in general, stresses and strains are only allowed within the basal plane, while the z -direction is strain/stress free (i.e., $\epsilon_3 = \epsilon_4 = \epsilon_5 = 0$ and $\sigma_3 = \sigma_4 = \sigma_5 = 0$). Such a condition nullifies e_{15}/d_{15} and e_{33}/d_{33} . As such, eq 3 is reduced to

Table 1. Piezoelectric Coefficients e_{11}/d_{11} and e_{31}/d_{31} of MXY Monolayers and e_{11}/d_{11} of MX₂ Monolayers^a

MXY	e_{11}	d_{11}	e_{31}	d_{31}	MX ₂	e_{11}	d_{11}
MoSSe	3.74	3.76	0.032	0.020	MoS ₂	3.56 (3.64, ^b 3.62 ^c)	3.34 (3.73, ^b 3.65 ^c)
MoSeTe	4.35	5.30	0.037	0.030	MoSe ₂	3.84 (3.92, ^b 3.83 ^c)	4.17 (4.72, ^b 4.55 ^c)
MoSTe	4.53	5.036	0.038	0.028	MoTe ₂	4.98 (5.43, ^b 4.67 ^c)	7.00 (9.13, ^b 7.39 ^c)
WSSe	2.57	2.26	0.018	0.011	WS ₂	2.46 (2.47, ^b 2.43 ^c)	2.02 (2.19, ^b 2.12 ^c)
WSeTe	3.34	3.52	0.010	0.008	WSe ₂	2.66 (2.7, ^b 2.57 ^c)	2.53 (2.79, ^b 2.64 ^c)
WSTe	3.48	3.33	0.010	0.007	WTe ₂	3.52 (3.40, ^b 3.23 ^c)	4.29 (4.60, ^b 4.39 ^c)

^aThe units of e_{ij} and d_{ij} are 10⁻¹⁰ C/m and pm/V, respectively. The ground state of the WTe₂ monolayer is the 1T phase, but its metastable 2H phase is considered here to be consistent with other MX₂ materials. ^bRef 5. ^cRef 9.

$$e_{ij} = \begin{pmatrix} e_{11} & -e_{11} & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & -0.5e_{11} \\ e_{31} & e_{31} & \cdot & \cdot & \cdot \end{pmatrix} \quad (7)$$

where there exist two independent piezoelectric coefficients, namely, e_{11} and e_{31} . Similarly, d_{ij} in eq 4 is reduced to

$$d_{ij} = \begin{pmatrix} d_{11} & -d_{11} & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & -2d_{11} \\ d_{31} & d_{31} & \cdot & \cdot & \cdot \end{pmatrix} \quad (8)$$

Here, d_{11} and d_{31} are derived by substituting eqs 7 and 8 into eq 5, which yields

$$d_{11} = e_{11}/(C_{11} - C_{12}) \quad (9)$$

and

$$d_{31} = e_{31}/(C_{11} + C_{12}) \quad (10)$$

The conventional MX₂ monolayers [$\bar{6}m2$ (D_{3h}) symmetry] possess a reflection symmetry with respect to the central M atomic plane. Such a condition requires that $e_{31} \equiv 0$ and $d_{31} \equiv 0$ in MX₂ monolayers, meaning that their piezoelectric polarizations are confined along the in-plane armchair direction. In the MXY monolayer structure (Figure 1a), the difference in atomic sizes and electronegativities of X and Y atoms gives rise to inequivalent M–X and M–Y bonding lengths and hence charge distributions. For example, in MoSTe, more charges are transferred from Mo to S than from Mo to Te (Figure 1b). This imbalance breaks the reflection symmetry along the vertical direction, resulting in a lower degree of 3m symmetry than the conventional MX₂ monolayers. Therefore, in the Janus MXY monolayers, both e_{11}/d_{11} and e_{31}/d_{31} are nonzero. In other words, both in-plane (along the armchair direction) and vertical piezoelectric polarizations are allowed in Janus MXY monolayers when they are subject to a uniaxial in-plane strain. When they are subject to biaxial in-plane strain, however, the in-plane piezoelectric polarization will be suppressed, while the out-of-plane one still will remain, according to eq 7.

In this study, we obtain e_{11} and e_{31} for MXY monolayers as well as e_{11} for MX₂ monolayers (Table 1) by fitting their piezoelectric polarizations as a function of a uniaxial in-plane strain ϵ_1 (along the armchair direction), which is varied from -0.5% to 0.5% with a step size of 0.1% (see Figure 1c as an example). This range of strain ensures that the piezoelectric effect stays in the linear regime as described by eq 1 and higher-order piezoelectric effects¹⁵ can be neglected. The values of d_{11} and d_{31} are calculated according to eqs 9 and 10, with the elastic coefficients C_{11} and C_{12} reported in the Supporting Information (Table S1). The calculated values of e_{11}/d_{11} of MX₂ monolayers (Table 1) are very close to the values reported in previous DFT

studies,^{5,9} confirming the validity of our method. The values of e_{11}/d_{11} of the Janus MXY monolayers fall between those of the MX₂ and MY₂ monolayers, as expected. From Table 1, we can infer the common trends regarding the relation between e_{11}/d_{11} and the choice of M and X/Y elements: First, the MoXY and MoX₂ monolayers have higher e_{11}/d_{11} values than WXY and WX₂ monolayers. Second, for a given metal element M, the monolayers containing heavier chalcogenide atoms have larger e_{11}/d_{11} values in general. As such, the MoTe₂ monolayer has the highest e_{11}/d_{11} values among the Janus and conventional monolayers in Table 1. It is also interesting to find that MoSTe and MoSeTe have similar e_{11}/d_{11} values, which are second highest in Table 1, because they contain a Te atom in the chalcogen site of their unit cells. Note that the magnitudes of d_{11} values reported in Table 1 ranging from 2.02 pm/V to 7.00 pm/V are comparable and even higher than commonly used 3D piezoelectric materials such as α -quartz ($d_{11} = 2.27$ pm/V)¹⁶ and AlN ($d_{33} = 5.6$ pm/V),¹⁷ implying the potential of the in-plane piezoelectric effects in these monolayer materials.

More significantly, the Janus MXY monolayers possess the vertical piezoelectric effect, characterized by e_{31}/d_{31} . Among the monolayers we studied, MoSTe and MoSeTe have the highest values of e_{31}/d_{31} (Table 1). We note that e_{31}/d_{31} in these materials are smaller by 2 orders of magnitude compared to e_{11}/d_{11} . However, the possibility to achieve vertical piezoelectric polarizations/fields in a 2D material is still highly desired, as they greatly increase the flexibility of piezoelectric device operations and their compatibility with existing microelectronic technologies [such as complementary metal–oxide–semiconductor (CMOS) transistors], where most of the functional layers are vertically stacked. Prior to our study, the only 2D materials that are identified to have a nontrivial vertical piezoelectric effect are III–V monolayers (e.g., AlAs and GaN) in their metastable buckled hexagonal 2D structures,⁹ which also lack a reflection symmetry along the vertical direction. Such vertical piezoelectric polarizations, however, are forbidden in their ground states, which have planar 2D structures similar to the 2D BN lattice. Furthermore, neither the ground states nor the metastable structures of these III–V monolayers have been synthesized, to the best of our knowledge. Therefore, it is not clear whether these buckled 2D III–V metastable structures can be realized in the near future. On the contrary, the materials systems we study, i.e., the Janus MXY monolayers, have already been fabricated using commonly used chemical vapor deposition techniques, with MoSSe being a successful example.¹³

We are glad to see that, during the preparation of this article, an experimental demonstration of piezoelectricity in the MXY Janus monolayers has been published.¹⁸ This demonstrates the ever-growing interest in the piezoelectricity of 2D materials. This paper, which considers piezoelectric properties along the

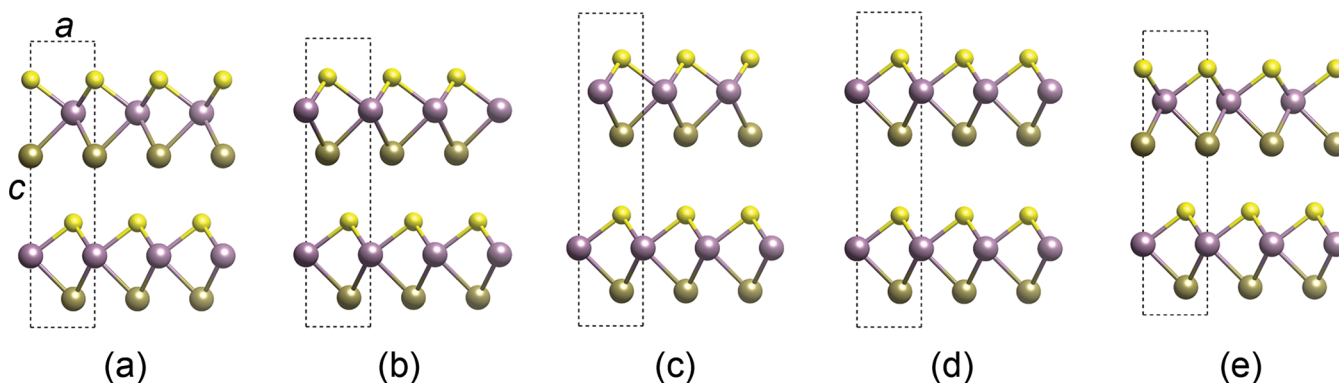


Figure 2. Multilayer MoSTe structures in five high-symmetry stacking sequences. Their unit cells, defined by lattice parameters a and c , are marked by a dashed line. The two constituent monolayers in the unit cells of (a), (b), and (c) have antiparallel orientations in the basal plane, whereas those in (d) and (e) have parallel orientations.

vertical direction, does not present a quantitative measurement of the piezoelectric coefficients. However, we expect follow-up studies to provide experimental values to compare with the theoretical calculations presented here.

MXy Multilayers. While we have demonstrated the viability to achieve vertical piezoelectric polarizations/fields in MXy monolayers, it is relatively weak compared with their in-plane counterparts. Here, we proceed to consider the piezoelectric properties of multilayer MXy structures, which can be constructed by stacking Janus TMD monolayers, thanks to their feasibility to be transferred using the standard mechanical exfoliation techniques for 2D materials.¹³ To ascertain the role of stacking sequences on the piezoelectric effects of the multilayer structures of Janus TMDs, we simulate the multilayer structures using bulk unit cells that are composed of two monolayers and consider five possible types of high-symmetry stacking sequences (Figure 2). A common feature of the five bulk unit cells is that the X atoms are always placed above the metal M atomic layer, while the Y atoms are always underneath. In other words, no monolayers are flipped along the vertical direction after the stacking. Such an arrangement ensures that the vertical piezoelectric effect will not be negated due to the flipping of one monolayer in the unit cells and can in principle be easily achieved in experiments. The unit cell in Figure 2a is analogous to the 2H stacking sequence of the conventional bulk MX₂ and, hence, is found to be lowest in energy among the five unit cells in Figure 2.

In these three-dimensional (3D) van der Waals crystals, e_{11} , e_{31} , e_{33} , and e_{15} are all distinct, and hence eqs 3 and 4 cannot be further reduced to describe the piezoelectric tensors. Note that d_{il} can be derived from e_{il} according to eqs 5 and 6, giving

$$d_{11} = e_{11}/(C_{11} - C_{12}) \quad (11)$$

$$d_{31} = (C_{33}e_{31} - C_{13}e_{33})/[(C_{11} + C_{12})C_{33} - 2C_{13}^2] \quad (12)$$

$$d_{33} = [(C_{11} + C_{12})e_{33} - 2C_{13}e_{31}]/[(C_{11} + C_{12})C_{33} - 2C_{13}^2] \quad (13)$$

and

$$d_{15} = e_{15}/C_{44} \quad (14)$$

The units of e_{il} , d_{il} , and C_{kl} in 3D crystals are C/m², pm/V, and 10¹⁰ Pa, respectively, different from the 2D monolayer case.

We first report the calculated values of d_{il} and e_{il} of the Janus TMD multilayers in the ground-state unit cell in Figure 2a. For this structure, the in-plane piezoelectric effect does not exist,

i.e., $e_{11} = 0$ C/m² and $d_{11} = 0 \times 10^{10}$ Pa, due to the fact that the two monolayers are antiparallel in the basal plane. An intuitive way to understand this point is that the two monolayers in such a unit cell always generate equal but opposite in-plane polarizations, which cancel each other. But the vertical electrical polarizations can still be induced by either in-plane or out-of-plane strains. Values of their e_{31}/d_{31} and e_{33}/d_{33} are listed in Table 2, and the related C_{ij} values are reported in Table S2 in

Table 2. Piezoelectric Coefficients e_{31}/d_{31} and e_{33}/d_{33} of the Janus MXy Multilayers with the Crystal Structure in Figure 2a^a

MXy	e_{31}	d_{31}	e_{33}	d_{33}
MoSSe	0.038	−0.109	0.352	5.248
MoSeTe	0.006	−0.467	0.428	6.217
MoSTe	0.119	−0.200	0.835	10.575
WSSe	0.044	−0.017	0.294	5.319
WSeTe	0.067	−0.031	0.374	6.710
WSTe	0.140	−0.002	0.752	9.279

^aThe units of e_{il} and d_{il} are C/m² and pm/V, respectively.

the Supporting Information. For a given MXy multilayer structure, the magnitude of e_{33} (d_{33}) is 1 order of magnitude higher than that of e_{31} (d_{31}), meaning that the vertical piezoelectric polarization due to a vertical strain is much stronger than in-plane strains. MoXY and WXY multilayers have similar values of e_{33}/d_{33} for a given combination of X and Y atoms; that is, the vertical piezoelectric effect relies little on the types of transition metal element. But e_{33}/d_{33} does depend strongly on the combination of X and Y atoms. From Table 2, e_{33}/d_{33} are always lowest in MSSe (M = Mo or W) but highest in MSTe (M = Mo or W). This is understandable since the difference in their atomic sizes breaks the reflection symmetry along the vertical direction and gives rise to vertical piezoelectric polarization. Therefore, it is not surprising to see a positive and almost linear correlation between e_{33}/d_{33} and a difference (Δr) in the radii of X and Y atoms (Figure 3), where the atomic radii of S, Se, and Te atoms are taken as 0.88, 1.03, and 1.23 Å, respectively.¹⁹ The multilayers with the highest d_{33} values are MoSTe ($d_{33} = 10.575$ pm/V) and WSTe ($d_{33} = 9.279$ pm/V), which are higher than d_{33} (=5.6 pm/V) of AlN.

To study the effect of interlayer coupling on the piezoelectric coefficients of MXy multilayers, we increased and decreased the vertical lattice parameter c of the ground-state structure of

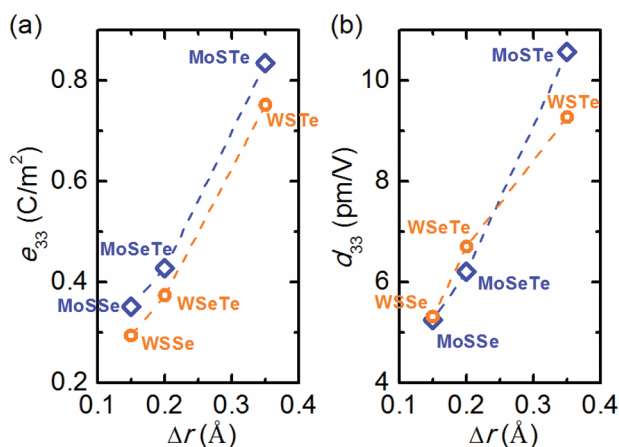


Figure 3. Correlation of (a) e_{33} and (b) d_{33} of multilayer MXY structure shown in Figure 2a with Δr , the difference in the atomic radii of X and Y atoms.

MoSeTe multilayers by 2%. Table S3 in the Supporting Information clearly shows that the out-of-plane piezoelectric coefficient e_{33} is enhanced when c is decreased (*i.e.*, when the interlayer interaction gets stronger), while e_{33} is decreased when c gets larger (*i.e.*, when the interlayer coupling gets weaker). On the contrary, changes in e_{31} are much smaller compared to the case of e_{33} . The above results clearly demonstrate the significant impact of interlayer coupling on piezoelectric effects in multilayer MXY structures. A rational understanding is that each MXY monolayer acts as a dipole moment, and the dipole–dipole interaction in multilayer structures modifies the charge distribution in each layer, which gives rise to a stronger piezoelectric effect.

Next, we take multilayer MoSeTe as an example to investigate the role of stacking sequences on the piezoelectric properties, especially on the magnitudes of e_{33}/d_{33} , of these Janus MXY TMDs. Their lattice parameters and piezoelectric coefficients are listed in Table 3, and their elastic coefficients are reported in Table S4 in the Supporting Information. From Table 3, the unit cells in Figure 2a–c have two constituent monolayers antiparallel in the basal plane, leading to $e_{11} = 0$ C/m² due to the reason mentioned earlier, while those in Figure 2d and e have two constituent monolayers parallel in the basal plane, allowing in-plane piezoelectricity. It is interesting to note that d_{11} of the multilayer MoSeTe unit cells in Figure 2d and e is ~ 4.95 pm/V (Table 3), very close to that of the MoSeTe monolayer (5.04 pm/V, see Table 1), showing that the interlayer van der Waals interactions have little effect on the in-plane piezoelectric polarization of each constituting monolayer when both are parallel in the basal plane.

Furthermore, the values of d_{33} vary with different stacking sequences between 5.700 pm/V (Figure 2d) and 13.517 pm/V (Figure 2b). From Table 3, we find an inverse correlation

between the d_{33} values of the five unit cells and their lattice parameter c or, equivalently, their interlayer distance. Note that c of the unit cells in Figure 2c and d are larger (~ 14 Å) because the S atoms in the lower monolayer and the Te atoms in the upper monolayer overlap along the vertical direction, and their values of d_{33} (and e_{33}) are lower. On the contrary, c of the unit cells in Figure 2a, b, and e are smaller (12.7–12.9 Å), and commensurately, the values of d_{33} (and e_{33}) are higher. Such a comparison clearly demonstrates that the interlayer van der Waals interactions may significantly affect the vertical piezoelectric effect in multilayer Janus MXY structures.

The above results have several important implications on the experimental measurements of the piezoelectric effect using the multilayer Janus MXY materials (bulk and thin films), where an arbitrary relative rotation angle between adjacent layers induces moiré patterns in layered van der Waals materials.²⁰ The in-plane piezoelectric polarization (due to a uniaxial in-plane strain) in these twisted multilayer structures may be much smaller than their constituent monolayers for two reasons. First, only a small amount of strain can be transferred from one layer to the other, usually not exceeding 1%, due to the weak van der Waals interlayer interactions.²¹ Second, the piezoelectric polarizations in the constituting monolayers may partially cancel each other because of their different orientations in the basal plane, as discussed earlier. Therefore, the multilayer structures are not suitable for operations based on the in-plane piezoelectric polarization (*i.e.*, e_{11}/d_{11}). However, a uniaxial strain (such as compression) along the vertical direction can still stimulate a very strong piezoelectric polarization/field in the same direction, as characterized by e_{33}/d_{33} in the multilayer MXY structures. Although our calculations are based on structures under five high-symmetry stacking sequences, the range of e_{33}/d_{33} we have obtained should also be appropriate for the randomly oriented multilayer structures in experiments. The reason is that an arbitrary relative angle between adjacent monolayers in experiments is at an intermediate state between two high-symmetry stacking sequences in our simulations. Furthermore, moiré patterns in multilayer TMD structures result in an unevenly distributed interlayer distance in various regions of the basal plane, but such variations have also been taken into account by the five unit cell structures in Figure 2, whose lattice parameter c changes by up to 1.3 Å in multilayer MoSeTe structures (see data in Table 3). Therefore, it is safe to expect that the measured d_{33} of this particular material falls in the range between 5.70 and 13.52 pm/V, which is higher than the conventional covalent 3D piezoelectric materials α -quartz ($d_{11} = 2.27$ pm/V)¹⁶ and AlN ($d_{33} = 5.6$ pm/V).¹⁷ Other multilayer Janus TMDs considered here also have values of d_{33} (Table 2) comparable to α -quartz and AlN.

The large values of d_{33} in the multilayer Janus TMD materials are highly desired for energy and electronic applications. For instance, they can potentially replace AlN in surface acoustic

Table 3. Lattice Parameters a and c (unit: Å), Piezoelectric Coefficients e_{ii} (unit: C/m²) and d_{ii} (unit: pm/V), and Relative Energy ΔE (unit: meV) of Multilayer MoSeTe Unit Cells with the Structures in Figure 2

structure	a	c	e_{11}	d_{11}	e_{31}	d_{31}	e_{33}	d_{33}	ΔE
(a)	3.318	12.764	/	/	0.119	−0.200	0.835	10.575	0
(b)	3.331	12.866	/	/	0.041	−1.234	0.666	13.517	90
(c)	3.318	13.972	/	/	0.075	−0.049	0.433	7.200	284
(d)	3.318	14.065	0.640	4.945	0.053	−0.022	0.332	5.700	296
(e)	3.325	12.701	0.680	4.958	0.104	−0.744	0.853	12.751	9

wave devices, sensors, and actuators,²² as van der Waals TMDs materials can be deposited or exfoliated with high crystal quality on various substrates including silicon, whereas deposition of AlN on these substrates usually requires a buffer layer due to lattice mismatch. Another benefit of utilizing these Janus TMD materials to replace AlN in such applications is the viability for device miniaturization. As a rough estimation, 100 layers of the Janus TMDs are only 65–70 nm thick, but AlN thin films at this thickness usually suffer from high defect densities (due to lattice mismatch with the substrate), and therefore they have to be thicker by 1 to 2 orders of magnitude to ensure high crystal quality. Other benefits of using multilayer TMD materials include compatibility with state-of-the-art 2D technologies, high breakdown current,²³ and a wide modulation range for the dielectric constant by applying an external electric field.²⁴ Finally, we note that d_{33} of the Janus multilayer TMDs is much smaller than lead zirconate titanate (PZT), the most commonly used ferroelectric/piezoelectric material in microelectromechanical systems (MEMS). However, PZT is known to contaminate silicon substrates, is toxic to the environment, and needs to be electrically poled for applications due to different domain orientations.²⁵ Therefore, the Janus multilayer TMDs may be an alternative low-cost and environmentally benign materials for MEMS applications.

Finally, the van der Waals multilayers have relatively small tensile strength along the out-of-plane direction, compared to conventional 3D materials. However, the peak strains in 3D piezoelectric devices are usually very small. For example, piezoelectric devices based on ZnO are usually operated with peak strains of 0.055% to 0.1%.²⁶ Given that the vertical piezoelectric effect in MXY multilayers are comparable and even stronger than ZnO and AlN, MXY multilayers may still find practical piezoelectric applications in the tensile strain regime. On the contrary, the load-carrying capability of MXY multilayers in compression is much larger, as the interlayer spacing can be reduced significantly. As such, it is better to operate the MXY multilayers with compressive strains. This makes them well suited for piezoelectric sensors, which are electromechanical systems that react to compression.

CONCLUSIONS

We have demonstrated strong piezoelectric effects in monolayer and multilayer Janus TMD materials with a general chemical formula of MXY (M = Mo, W; X/Y = S, Se, and Te) through first-principles simulations. For MXY monolayers, a strong in-plane piezoelectric polarization is induced by a uniaxial strain in the basal plane characterized by the piezoelectric coefficients e_{11}/d_{11} , similar to the phenomena previously observed in conventional MX₂ monolayers (e.g., MoS₂).^{5–7,9} Different from prior studies, however, we also report a vertical piezoelectric polarization in the MXY monolayers upon application of uniaxial or biaxial strains, characterized by e_{31}/d_{31} , due to the lack of reflection symmetry with respect to the central M atoms. This discovery inspired us to go on to investigate the piezoelectric effects in multilayer MXY materials. We find a strong piezoelectric polarization due to a vertical strain, characterized by e_{33}/d_{33} in multilayer Janus TMDs. The values of their d_{33} , although varying by a factor of 2 for different interlayer stacking sequences, are found to be comparable to or even superior than conventional 3D piezoelectric materials such as α -quartz and AlN. These results suggest the possibility of employing piezoelectric effects along the vertical direction in multilayer MXY materials for

electronics and energy applications. Considering that an example of the Janus MXY materials—MoSSe—has just recently been achieved,¹³ we hope our study will trigger more research interest into this type of TMD materials, especially for its applications in energy harvesting, sensors, actuators, and surface acoustic waves.

METHODS

Our DFT simulations are carried out using the Vienna ab initio simulation package (VASP).²⁷ Projector-augmented wave pseudopotentials²⁸ are used with a cutoff energy of 520 eV for plane-wave expansions. The exchange–correlation functional is treated within the Perdew–Burke–Ernzerhof (PBE) generalized gradient approximations (GGA).²⁹ Γ -Centered k -point meshes of $18 \times 18 \times 1$ and $18 \times 18 \times 4$ in the first Brillouin zone are found to yield well-converged results for the unit cells of monolayer and multilayer structures, respectively, during the structural relaxations. A 16 Å thick vacuum space was found to be large enough to prevent any interactions between the adjacent periodic images of the 2D monolayers. The long-range van der Waals interactions in multilayer MXY crystals are treated using the DFT-D3 method developed by Grimme *et al.*^{30,31} The atomic positions of the unit cells are optimized until all components of the forces on each atom are reduced to values below 0.01 eV/Å. The in-plane lattice parameter a of a given MXY monolayer is between that of MX₂ and MY₂, as expected (Table S1 in the Supporting Information). The (relaxed-ion) elastic stiffness coefficients are calculated using the finite difference method,³² and the electric polarizations are calculated using the Berry phase method as implemented in VASP.³³

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acsnano.7b03313.

Lattice parameters and elastic stiffness coefficients of MXY and MX₂ monolayers and MXY multilayer structures and the impact of interlayer coupling on the piezoelectricity of MXY multilayers (PDF)

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Notes

The authors declare no competing financial interest.

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